

Akhilesh Malish

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Professional Summary:

Experienced Software Developer with around 7 years of experience delivering full-stack web applications that are responsive, scalable, and maintainable. Well-versed in modern front-end frameworks like React.js, with additional exposure to Angular and Vue.js. Strong programming background with hands-on experience in Python, Java, and C++, combined with a deep understanding of core CS concepts like data structures and algorithms. Experienced in working with both relational and non-relational databases, applying Agile practices, and collaborating closely with cross-functional teams. Comfortable working with Git for version control, debugging complex issues, and ensuring code quality through reviews and testing. Familiar with deploying to cloud platforms like AWS and integrating CI/CD workflows to streamline releases.

Frontend Development (React.js, Angular, Vue.js):

- Developed dynamic and responsive web applications over the years using **React.js** as the primary framework, with hands-on experience also in **Angular** and **Vue.js** for client projects and internal tools.
- Built scalable user interfaces with a component-driven approach using **React.js**, **TypeScript**, **Redux Toolkit**, and **Tailwind CSS**, leveraging client-side routing (**React Router**) and performance optimizations like **code splitting** and **lazy loading**.
- Integrated **RESTful** and **GraphQL** APIs securely with **Axios** and **Apollo Client**, implementing authentication flows using **JWT**, **OAuth**, and **HttpOnly cookies** across different frameworks.
- Developed complex, validated forms using **React Hook Form**, **Formik**, and **Zod**, ensuring accessibility and responsiveness across devices using utility-first design with **Tailwind CSS**.
- Contributed to **component libraries** and **design systems**, using **Storybook** for component documentation and reuse across **React**, **Angular**, and **Vue** projects.
- Applied **React Hooks** extensively and refactored legacy **class components** into clean, **functional components**; utilized state management tools like **Redux Toolkit** and **React Query** for efficient data handling.
- Created interactive dashboards and features using third-party libraries such as **Chart.js**, **react-select**, and **react-beautiful-dnd**, while maintaining performance through **virtualization** (e.g., **react-window**) and **memoization** techniques (**React.memo**, **useMemo**).
- Ensured quality and reliability with testing frameworks like **Jest**, **React Testing Library**, and **Cypress**, and followed **Agile** development practices with **Git**-based version control and **peer code reviews**.
- Delivered consistent cross-platform **UI/UX** by implementing **dark/light theme toggles**, persisting preferences using **Context API**, **localStorage**, and **sessionStorage**.
- Adapted best practices from each framework (**React**, **Angular**, **Vue**) to meet project needs, demonstrating flexibility and depth in **frontend architecture** and **user experience optimization**.

Backend Development (Python (FastAPI, Flask), Java (Springboot), C++):

- Developed and maintained robust backend services using **Python** with **FastAPI**, and **Java** with **Spring Boot**, ensuring scalable and maintainable RESTful API architectures across multiple projects.
- Leveraged **Python's async features** and **FastAPI's** automatic **OpenAPI/Swagger** documentation for high-performance request handling and developer-friendly API integration.
- Built secure backend logic using **JWT**, **OAuth2.0**, and **HttpOnly cookies**, ensuring proper access control and session management across web and mobile platforms.

- Designed efficient data access layers connected to **PostgreSQL**, **MySQL**, **MongoDB**, and **DynamoDB**, using ORMs like **SQLAlchemy**, **Tortoise ORM**, and **JPA/Hibernate** (Java), and async libraries like **motor**.
- Utilized **Pydantic** in Python for strict request/response validation and clean data modeling; followed type-safe practices in Java for robust compile-time checks.
- Built high-performance utilities and core logic in **C++**, using the **Standard Template Library (STL)** for efficient data structures and algorithms in compute-heavy modules.
- Employed background tasks and dependency injection in **FastAPI** and **Spring**, implementing features like scheduled jobs, email notifications, and file processing workflows.
- Developed **GraphQL APIs** using **Strawberry (Python)** and integrated with various data sources for dynamic and flexible client queries.
- Ensured backend quality and stability through automated unit/integration tests using **pytest**, **httpx**, **JUnit**, and **Mockito**, following TDD principles.
- Used **Docker** for containerization and deployed services on **AWS (EC2, Lambda)** and **GCP (Cloud Run)**, integrating CI/CD pipelines for continuous delivery.
- Collaborated closely with frontend teams (React, Angular, Vue) to define clear **API contracts**, maintain versioning, and resolve integration issues quickly.

Monitoring & Performance Optimization:

- Utilized tools like **AWS CloudWatch** to monitor software performance and health metrics, setting up **automated alerts** to proactively detect and resolve issues during development and production.
- Performed **debugging**, **profiling**, and **load testing** to ensure system reliability under high concurrency, optimizing backend services written in **Java**, **Python**, and **C++**.
- Implemented **caching strategies** using **Redis** and leveraged **CDN integrations** to improve response time and reduce server load for frequently accessed resources.
- Conducted regular **code reviews** and refactoring sessions focused on performance tuning and memory management across services.

Cloud-Based Web Application Architecture:

- Designed and implemented **cloud-native applications** using **AWS** services including **EC2**, **S3**, **Lambda**, **IAM**, **CloudWatch**, and **RDS**, enabling scalable, fault-tolerant architectures.
- Integrated **React.js**, **Angular**, and **Vue.js** frontends with **Node.js**, **FastAPI (Python)**, and **Java Spring Boot** backends to build full-stack solutions aligned with business requirements.
- Developed **secure, RESTful APIs** and **microservices**, ensuring seamless deployment pipelines through **CI/CD** and infrastructure as code.
- Supported both **relational (PostgreSQL, MySQL)** and **non-relational (MongoDB, DynamoDB)** databases in cloud environments, enabling high availability and horizontal scalability.
- Built reusable backend components and automation scripts using **Python** to streamline deployment and system management processes.

Project Highlights:

- Developed and maintained scalable front-end features using **React.js**, **Angular**, and **Vue.js**, with a strong focus on component-driven architecture, code reusability, and performance optimization.
- Translated UI/UX designs into responsive, accessible web interfaces using **Tailwind CSS**, **Material UI**, and **Bootstrap**, ensuring cross-browser and cross-device compatibility.

- Integrated **RESTful** and **GraphQL** APIs built with **Python (FastAPI)**, **Java**, and **C++**, utilizing tools like **Axios**, **React Query**, and **Apollo Client** for efficient data fetching and state management.
- Built full-stack solutions with structured communication between frontend and backend services, maintaining clean API contracts and secure authentication with **JWT**, **OAuth2**, and **HttpOnly cookies**.
- Worked with both **SQL** (PostgreSQL, MySQL) and **NoSQL** (MongoDB, DynamoDB) databases to support robust data flows in enterprise-level applications.
- Ensured accessibility compliance (WCAG 2.1 AA) using semantic HTML, ARIA roles, and keyboard navigation, delivering inclusive experiences to all users.
- Diagnosed and optimized performance issues using tools like **Chrome DevTools**, **React Profiler**, and **Lighthouse**, improving app load times and responsiveness.
- Established **CI/CD pipelines** with **GitHub Actions**, **Jenkins**, and **GitLab CI**, automating testing and deployment processes on cloud platforms including **AWS** and **Azure**.
- Maintained high-quality standards through unit and integration testing with **Jest**, **React Testing Library**, and **Cypress**, supporting continuous delivery.
- Actively contributed to Agile processes by participating in sprint planning, code reviews, and daily standups in cross-functional development teams.

TECHNICAL SKILLS:

Languages	JavaScript, TypeScript, Python, SQL, PL/SQL, Shell Scripting, Python, C, C++
Front-end	React.js, Angular, Vue.js, Next.js, TypeScript, Redux Toolkit, React Query (TanStack), Zustand, React Context API, Tailwind CSS, Bootstrap, Storybook, React Testing Library, Jest
Back-end	Python (FastAPI, Flask), Java (Spring Boot), Node.js, Express.js, RESTful API Development, GraphQL, Microservices Architecture, Event-Driven Design, PostgreSQL, MySQL, MongoDB, Query Optimization, Connection Pooling
State Management	Redux Toolkit, Redux Thunk, React Query (TanStack), React Context API, Zustand
Monitoring & Visibility	Prometheus, Grafana, CloudWatch
Devops	Docker, Kubernetes (EKS), Jenkins, GitHub Actions, Terraform, CloudFormation, CI/CD pipelines, Serverless (Cloudflare)

Education:

Bachelor's in Computer Science

Master's in Computer Science

Professional Experience:

Pace Solutions (Sam's Club) – Reston, VA
Sr. Software Engineer

May 2024 -Present

Project:

Led the frontend development of a scalable, modular web platform delivering multiple business applications using **React.js**, **TypeScript**, and **Tailwind CSS**, integrated with backend services in a cloud-native AWS environment.

Responsibilities:

- Developed **responsive, accessible, and high-performance user interfaces** with **React.js**, leveraging **functional components** and **React Hooks** to build clean, maintainable code.
- Managed complex application state using **Redux Toolkit**, **React Context API**, and **Zustand**, ensuring efficient data flow and predictable state management.
- Built reusable UI components following **component-driven architecture** principles and maintained a consistent design system with **Storybook** for rapid development and collaboration.
- Integrated RESTful APIs with **Axios**, implementing advanced patterns like **caching**, **optimistic updates**, and **pagination** for smooth data handling.
- Applied **form management and validation** using **React Hook Form**, **Formik**, and **Zod** to handle complex forms with ease and ensure robust input validation.
- Enhanced performance through **code splitting**, **lazy loading**, and **React Suspense**, optimizing load times and user experience.
- Ensured accessibility compliance with **WCAG 2.1 AA** standards, using semantic HTML and ARIA roles for inclusive user experiences.
- Collaborated with cross-functional teams to translate **Figma designs** into pixel-perfect, mobile-first interfaces using **Tailwind CSS** and **Bootstrap** utilities.
- Authored **unit and integration tests** with **Jest** and **React Testing Library**, integrating tests into automated CI pipelines for reliable code delivery.
- Implemented monitoring and logging of frontend applications using **AWS CloudWatch**, supporting proactive issue detection.
- Participated in Agile ceremonies, working closely with product managers, designers, and backend engineers to deliver business-driven features.

Environment:

React.js, TypeScript, Redux Toolkit, Zustand, React Context API, React Query, Axios, Tailwind CSS, Bootstrap, Storybook, React Hook Form, Formik, Zod, Jest, React Testing Library, AWS (CloudWatch, S3, CloudFront), GitHub Actions, Jenkins, Figma, Jira, Confluence

Tata Consultancy Services – Hyderabad, India
Software Engineer

Jul 2021-Dec 2022

Project:

Worked as a full-stack developer on a data-driven business analytics platform, delivering responsive, scalable frontends using **Angular** integrated with **Java Spring Boot** backend services in a cloud-native environment.

Responsibilities:

- Developed reusable, modular UI components and responsive layouts using **Angular**, **TypeScript**, and **RxJS**, focusing on maintainability and performance.
- Managed complex application state with **NgRx** and Angular services, ensuring predictable and scalable state management.
- Built dynamic, form-heavy user interfaces using **Angular Reactive Forms** with **custom validators** to improve form validation and user experience.
- Integrated RESTful APIs with Angular's **HttpClient**, handling error management, interceptors for authentication tokens, and efficient data fetching using **RxJS operators**.
- Collaborated closely with backend engineers to design and consume secure **Java Spring Boot REST APIs**, including JWT-based authentication and OAuth2.0 mechanisms.
- Implemented session management and security best practices with **HttpOnly cookies** and OAuth2.0 flows in coordination with Spring Security.
- Translated UI/UX wireframes and designs from **Figma** into pixel-perfect, mobile-first Angular components with **Angular Material** and **Bootstrap**.
- Ensured accessibility compliance following **WCAG 2.1 AA** standards using semantic HTML, ARIA attributes, and keyboard navigation support.
- Wrote unit and integration tests using **Jasmine**, **Karma**, and **Angular Testing Utilities** for frontend reliability; contributed to backend tests with **JUnit** and **Mockito**.
- Automated build, test, and deployment processes using **Jenkins** and **GitHub Actions** within CI/CD pipelines for Angular and Spring Boot applications.
- Participated in Agile development processes, collaborating with cross-functional teams including UI/UX designers, backend developers, and QA engineers.

Environment:

Angular, TypeScript, RxJS, NgRx, Angular Reactive Forms, Angular Material, Bootstrap, Java, Spring Boot, Spring Security, RESTful APIs, JWT, OAuth2.0, HttpClient, Jasmine, Karma, JUnit, Mockito, Jenkins, GitHub Actions, Docker, AWS (S3, EC2, RDS, IAM, CloudWatch), Figma, Jira, Confluence

ESH Solutions – Jersey Channel Islands, UK (Remote)
Software Engineer

Oct 2020 - Jun 2021

Project:

Led the front-end development of a modular analytics dashboard platform, creating scalable and reusable components using **Vue.js**, **Vuex**, and **Tailwind CSS**, optimized for performance and usability, integrated with Python-based backend services.

Responsibilities:

- Designed and implemented a microfrontend architecture using **Webpack Module Federation**, enabling scalable and independent front-end deployments.
- Developed complex, high-performance Vue.js components for interactive dashboards, using **Vuex** for robust state management and **Vue Router** for navigation.
- Integrated **Chart.js** for dynamic data visualization, delivering real-time insights through line, bar, and donut charts.
- Applied lazy loading, component-level caching, and code splitting to improve application responsiveness and reduce bundle size.

- Implemented role-based access control (RBAC) and feature flags to deliver secure and customizable UI experiences.
- Styled responsive UI elements with **Tailwind CSS** and ensured design consistency using reusable Vue components.
- Documented reusable UI components in **Storybook**, streamlining development and onboarding across teams.
- Collaborated with DevOps teams to support CI/CD pipelines using Docker, GitHub Actions, and Kubernetes for seamless deployment.
- Worked closely with backend engineers to integrate **Python Flask/FastAPI** RESTful APIs, ensuring smooth data exchange and authentication workflows.
- Ensured WCAG-compliant, accessible user interfaces through collaboration with designers and stakeholders, achieving pixel-perfect UI implementations.

Environment:

Vue.js, Vuex, Vue Router, TypeScript, Tailwind CSS, Chart.js, Storybook, Python (Flask, FastAPI), RESTful APIs, Webpack (Module Federation), Docker, GitHub Actions, Kubernetes, AWS (Lambda, EC2), Jira, Confluence

Lloyds Bank (ESH Solution) UK (Remote)
Associate Software Engineer

Feb 2018 – Sept 2020

Project:

Contributed to the development and maintenance of a full-stack inventory management system, delivering responsive and interactive React.js frontends for warehouse and retail clients. Enabled real-time product tracking, dynamic dashboards, and streamlined inventory management.

Responsibilities:

- Developed and maintained React.js-based admin dashboards with reusable, modular components enabling CRUD operations, stock alerts, and real-time inventory visualization through tables, filters, and search functionality.
- Used React Hooks and Context API for efficient state management and side effects, ensuring a smooth user experience.
- Integrated **Chart.js** within React components to display real-time inventory trends, supplier performance, and restock cycles with interactive and responsive charts.
- Collaborated with backend teams to consume RESTful APIs built with Node.js/Express, ensuring seamless and reliable data flow via Axios with proper error handling and loading states.
- Implemented client-side input validation using **Zod** schemas, complementing backend validation for robust and predictable data handling.
- Enhanced UI responsiveness and accessibility following best practices for keyboard navigation and ARIA attributes.
- Built role-based UI features to support different user roles such as admins, warehouse managers, and viewers, ensuring secure and appropriate access.
- Utilized Docker for containerized development environments and worked closely with DevOps teams on automated CI/CD pipelines using GitHub Actions.
- Participated in agile ceremonies including sprint planning, code reviews, and daily standups, using Jira, Confluence, and Slack to maintain team communication.

- Authored and maintained clear technical documentation for frontend components, API integration, and setup workflows.

Environment:

React.js, React Hooks, Context API, TypeScript, Axios, Zod, Chart.js, Node.js, Express.js, MongoDB, Docker, GitHub Actions, JWT, REST APIs, CI/CD, Jira, Confluence, Slack